THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

AMTI – NMTC – **25th October, 2026** – PRIMARY FINAL

Timings: 10am to 1pm

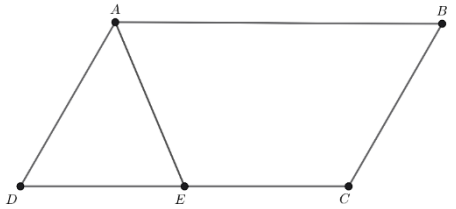
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Instructions:

1. There are 8 questions, answer all questions. Each question carries 10 marks.
2. Elegant and innovative solutions will get extra marks.
3. Diagram and justification should be given wherever necessary.
4. Before start answering, fill in the FACE SLIP completely.
5. Your **Rough Work** should be in the answer sheet itself.
6. Diagrams given here are only visual aids; they are NOT drawn to scale.

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1. ABCD is a parallelogram. The measure of ∠ABC = 60°. The bisector of ∠BAD (dividing ∠BAD in two equal halves) meets the side DC in point E such that E is the midpoint of side DC. If the length of segment AE = 6*cm*, then find: Perimeter of ΔADE + perimeter of □ABCE.



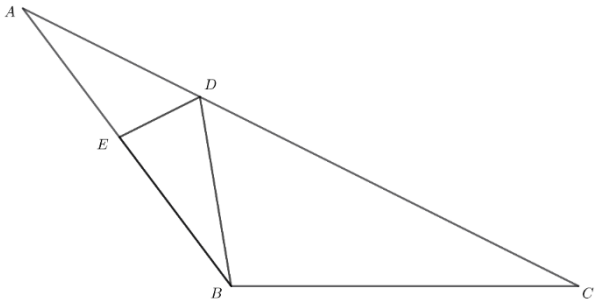
2. Form the smallest seven-digit number, completely divisible by 99, by using the digits 0, 2, 4, 6, 8 each at least once. Give the detailed justification.

3. In the following *Alphametic addition problem*, each letter represents a unique digit. It is known that when the digit represented by A is even and the digit represented by N is odd, then the *Alphametic addition problem* has two different solutions. Find these two solutions.

$$\begin{array}{cccccc} & A & M & T & I \\ + & N & M & T & C \\ \hline F & I & N & A & L \end{array}$$

4. If $\frac{1}{y}-\frac{1}{x}=7$, then evaluate: $\left(\frac{11x+5xy-11y}{5x-33xy-5y}\right)\left(\frac{35x+26xy-35y}{3x-20xy-3y}\right)$.

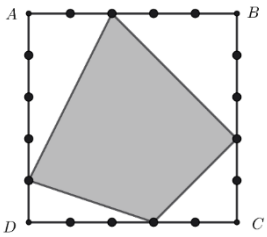
5. In ΔABC, AB = BC. Point D is on side AC and point E is on side AB such that BD = BE = AD. It is known that ∠ADE = 30°. Then find the measure of ∠ABC, ∠ACB, ∠BDC, ∠DEA.



6. Call a number *prime looking* if it is composite but not divisible by 2, 3, or 5. The three smallest *prime-looking* numbers are 49, 77, and 91. There are 95 prime numbers less than 500. How many *prime-looking* numbers are there less than 500?

7. Raj is to select six chocolates from a tray containing the chocolates of type (i) fruit & nut, (ii) roasted almond, (iii) milky-bar, (iv) blueberry. All chocolates of each type are identical. There are at least six chocolates of each of these four types in the tray. How many different assortments of six chocolates can be selected?

8. Each side of square ABCD with side length of 5*cm* is divided into equal five parts by four points. Choose one of the four points from each side, and connect the points consecutively to obtain a quadrilateral (One example is shown in the adjoining figure). Write all possible areas of the quadrilateral so obtained.



------x------x------x------